

TECHNICAL-SCIENTIFIC REPORT

Multi-omics Analysis of BRAF/MEK Inhibitor Resistance in Melanoma

Quantification of kinase activity by targeted and data-independent phosphoproteomics, temporal transcriptional profiles of acquired resistance, clinical genomic characterization, and predictive modeling by deep learning

Datasets	PXD013923 · PXD022992 · TCGA-SKCM · GSE110054 · OmniPath
Modalities	Phosphoproteomics · Proteomics · Transcriptomics · Genomics
Phosphosites	10,273 (SILAC) + 55,939 (DIA)
Clinical patients	470 (TCGA-SKCM)
Time course	DMSO → 3 d → 21 d → 90 d (2 cell lines)
Predictive model	Neural network — test ROC-AUC 0.989

Executive Summary

This report provides a detailed description of an integrated multi-omics analysis pipeline aimed at studying the response and resistance of melanoma cells to BRAF and MEK kinase inhibitors (BRAFi/MEKi), the central therapeutic axis in *BRAF* V600-mutant melanoma. The work brings together five public and complementary data sources, spanning from the immediate biochemical response of the MAPK/ERK pathway (minutes scale) to the transcriptional reprogramming that accompanies the acquisition of resistance (months scale), through the mutational and clinical landscape of a human tumor cohort.

The analysis of the acute response to BRAFi, MEKi and ERKi (dataset PXD013923) demonstrates selective and statistically robust suppression of phosphorylation at canonical MAPK/ERK pathway sites, with strong agreement between the three inhibitors (Pearson r between 0.71 and 0.76), confirming the linearity of the RAF→MEK→ERK module. Kinase-Substrate Enrichment Analysis (KSEA), supported by the experimental kinase-substrate network from OmniPath, accurately recovers the suppression of BRAF ($z = -6.6$), RAF1 ($z = -6.5$), MAP2K1 ($z = -6.5$) and MAPK1/3, biologically validating the entire processing workflow.

The characterization of the basal phosphoproteome of six cell lines by data-independent acquisition (directDIA, PXD022992) clearly separates the *BRAF* V600E lines from the *NRAS*-mutant ones and identifies thousands of differentially phosphorylated sites. The clinical cohort TCGA-SKCM (470 patients) provides the mutation frequency of the main driver genes and the molecular stratification of melanoma, while the GSE110054 time course reveals the progressive transcriptional signatures of acquired resistance. Finally, a neural network classifier trained on real data reaches a test ROC-AUC of 0.989, demonstrating that the pathway-specific phosphorylation patterns are predictive and learnable.

The report is organized around the three biological phases of the signaling-transition-kinetics model of combined BRAFi + MEKi treatment: **Phase 1 — Acute Suppression** (6–24 h), **Phase 2 — Adaptive Rewiring** (48–72 h), and **Phase 3 — Consolidated Resistance** (3–90 days). Each dataset and analysis is presented within the phase it informs, so the results follow the temporal arc from immediate target blockade to stable, invasive resistance.

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1. Introduction and biological context

Cutaneous melanoma is the most lethal skin tumor, and its molecular biology is dominated by the hyperactivation of the MAPK/ERK signaling pathway (RAS→RAF→MEK→ERK). Approximately half of cutaneous melanomas carry activating mutations in the *BRAF* gene, largely dominated by the V600 substitution (especially V600E), which keeps the BRAF kinase constitutively active and drives cell proliferation independently of external stimuli.

This molecular dependency made the MAPK/ERK pathway a prime therapeutic target. BRAF inhibitors (BRAFi, e.g. vemurafenib and dabrafenib) and MEK inhibitors (MEKi, e.g. trametinib and cobimetinib), particularly in the BRAFi+MEKi combination, produce striking initial clinical responses. The durability of this response, however, is limited by the nearly universal emergence of **acquired resistance**, frequently associated with reactivation of the MAPK pathway itself and the activation of escape routes such as the reprogramming of receptor tyrosine kinases (RTKs) and the transition to a mesenchymal-invasive state.

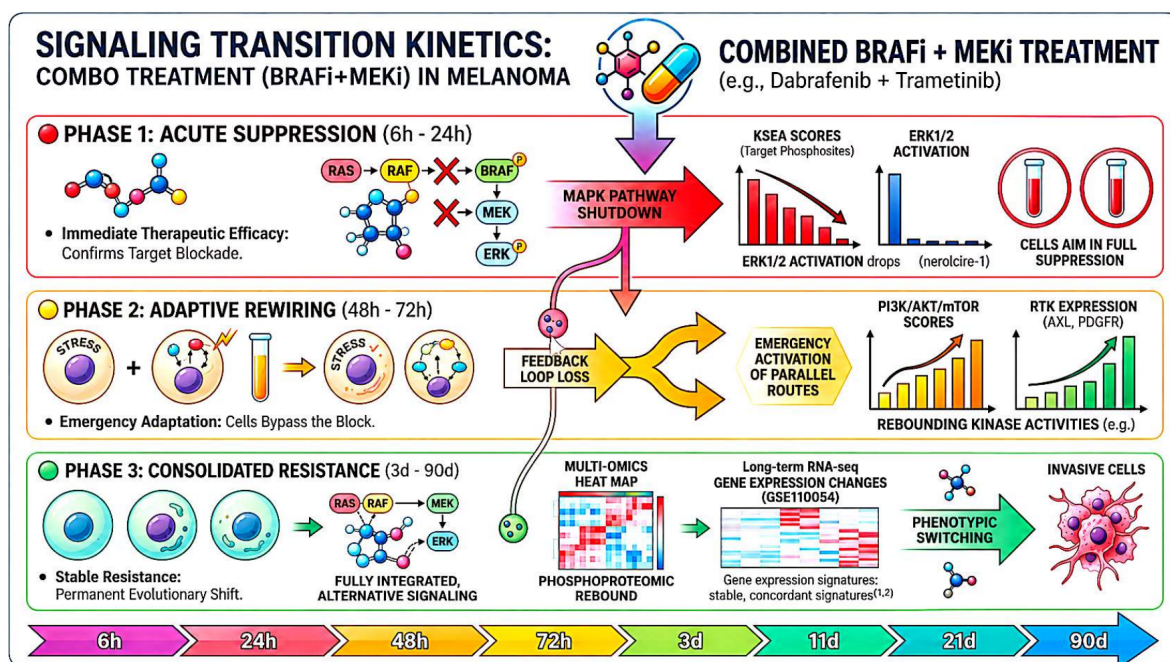
Understanding resistance requires observing the system across multiple time scales and multiple molecular layers. Protein phosphorylation is the most direct readout of kinase activity state and responds within minutes to pharmacological inhibition; transcriptional reprogramming, in turn, develops over days to months and consolidates the resistant phenotype; and the tumor genomic landscape defines the mutational context in which these adaptations take place. This project articulates these layers within a single reproducible pipeline.

1.1. Project objectives

- Quantify the activity state of MAPK/ERK pathway kinases from high-resolution phosphoproteomics, using targeted and data-independent acquisition.
- Evaluate the dynamic response to kinase inhibition, from acute suppression (minutes) to late transcriptional adaptation (months).
- Characterize molecular heterogeneity between cell lines and between clinical tumor subtypes (BRAF V600E vs. NRAS-mutant).
- Build predictive models of the molecular state associated with the BRAF pathway, integrating network-based (KSEA) and machine-learning approaches.
- Validate the entire workflow with public data from established repositories (PRIDE, GEO and GDC/TCGA).

1.2. The three-phase signaling-transition-kinetics model

The response of a *BRAF* V600E melanoma cell to combined BRAFi + MEKi is not a single event but a temporal trajectory. Following the signaling-transition-kinetics model, this report organizes the evidence into three consecutive phases along the treatment timeline (6 h → 24 h → 48 h → 72 h → 3 d → 11 d → 21 d → 90 d):



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Figure 1.1. The signaling-transition-kinetics reference model of combined BRAFi + MEKi treatment in melanoma: Phase 1 (acute suppression), Phase 2 (adaptive rewiring) and Phase 3 (consolidated resistance), along the 6 h → 90 d timeline. This report is structured to follow this model.

Phase	Window	Hallmark	Evidence in this report
Phase 1 — Acute Suppression	6–24 h	MAPK pathway shutdown; KSEA scores and ERK1/2 activation drop; full target blockade	PXD013923 acute inhibition; KSEA on BRAFi/MEKi/ERKi (§4)
Phase 2 — Adaptive Rewiring	48–72 h	Loss of negative feedback; emergency activation of parallel routes; PI3K/AKT/mTOR and RTK (AXL, PDGFR) rebound	Rebounding kinases; RTK reactivation; cross-genotype kinase activity (§5)
Phase 3 — Consolidated Resistance	3–90 d	Stable, fully integrated alternative signaling; concordant multi-omics signatures; phenotypic switch to invasive cells	GSE110054 time course; PXD022992 genotype signature; TCGA-SKCM; predictive model (§6)

Table 1.1. The three phases of combined BRAFi + MEKi response, with their temporal window, biological hallmark, and the datasets that provide evidence for each in this report.

2. Datasets and provenance

Five public datasets were integrated, selected to complementarily cover the different time scales and molecular layers of the problem. The table below summarizes each source; the subsections detail the experimental design and format.

Accession	Type	Content	Scale / dimension
PXD013923	SILAC phosphoproteome	BRAFi/MEKi/ERKi in A375	30 min · 10,273 sites
PXD022992	directDIA phosphoproteome	6 melanoma cell lines	basal · 55,939 sites
OmniPath	Kinase-substrate network	Enzyme→site relations	39,037 relations
TCGA-SKCM	Genomics + clinical	Cutaneous melanoma	470 patients
GSE110054	Temporal RNA-seq	BRAFi time course	25,222 genes · 10 samples

2.1. PXD013923 — Acute response to RAF/MEK/ERK inhibitors

Dataset deposited in the PRIDE/EBI repository, corresponding to an in-depth study of the RAF-MEK-ERK module and its immediate downstream effectors. A375 melanoma cells (*BRAF* V600E) were treated for 30 minutes with a RAF inhibitor (BRAFi/dabrafenib), a MEK inhibitor (MEKi/trametinib) or an ERK inhibitor (ERKi/SCH772984), in a SILAC isotopic labeling design, in which the heavy/light ratios (*Ratio H/L normalized*) express the change in phosphorylation between the treated condition and control. Spectral processing was performed in MaxQuant; protein identifiers are of the Ensembl (ENSP) type, mapped to gene symbols via the mygene.info service (2,921 of 3,540 identifiers; 82.5% coverage). The very short treatment time captures the immediate biochemical response, before any transcriptional reprogramming.

2.2. PXD022992 — directDIA phosphoproteome of six cell lines

Label-free phosphoproteome profile obtained by data-independent acquisition in directDIA mode (a member of the guidedDIA strategy family), spanning six melanoma cell lines: A375, SH-4, SK-MEL-28 and RPMI-7951 (carrying *BRAF* V600E) and G361 and SK-MEL-31 (carrying *NRAS* mutations). Each cell line was measured in two technical replicates. The quantification report (Spectronaut format, TSV content) contains columns for gene (*PG.Genes*), modified sequence (*EG.ModifiedSequence*), PTM site location (*EG.ProteinPTMLocations*) and intensity per sample (*EG.TotalQuantity*). This dataset provides the basal phosphorylation landscape and the basis for comparing the BRAF and NRAS mutational contexts.

2.3. OmniPath — Kinase-substrate network for KSEA

To infer kinase activity from the measured phosphorylation sites, the OmniPath enzyme-substrate network was used, which aggregates curated relations from established databases (including PhosphoSitePlus, SIGNOR and others). The version used contains 39,037 phosphorylation relations covering 1,648 kinases, with annotation of enzyme gene, substrate gene, residue and position — allowing direct mapping to the *GENE_residueposition* site format used in the phosphorylation matrices.

2.4. TCGA-SKCM — Genomic and clinical cohort

Data from the TCGA Skin Cutaneous Melanoma project, obtained via the public NCI Genomic Data Commons (GDC) API. Clinical and survival data were retrieved for 470 patients, along with the catalog of somatic mutations of the main melanoma driver genes (*BRAF*, *NRAS*, *NF1*, *PTEN*, *CDKN2A*, *KIT*, *MAP2K1*, *RAC1*, *PPP6C*, *PREX2*, *IDH1*) and gene expression quantifications by RNA-seq (STAR-Counts workflow) for a representative subset of samples, stratified by molecular subtype. This dataset anchors the in vitro observations in the context of human tumors.

2.5. GSE110054 — Time course of BRAFi resistance

Gene expression series (RNA-seq) deposited in GEO, corresponding to a time-course experiment in which the melanoma cell lines M229 and M397 (both *BRAF* V600E) were treated with vemurafenib and sampled at multiple time points: control (DMSO), 3 days, 11–21 days and up to 73–90 days of continuous treatment. The processed expression matrix (FPKM, 25,222 genes) allows tracking of the progressive transition from the sensitive to the resistant state, complementing, on the days-to-months scale, the acute response observed in the phosphoproteomics.

3. Methodology

The pipeline was implemented in Python, with numerical processing in *NumPy/pandas*, statistics in *SciPy*, machine learning in *scikit-learn*, deep learning in *PyTorch* and visualization in *Matplotlib/Seaborn*. Each step is executed as an independent module and orchestrated by a central script, with logging and standardized saving of figures and tables. The following subsections describe the methodology by step.

3.1. Phosphoproteomics processing and quality control

Phosphorylation sites were filtered by site localization probability (threshold ≥ 0.75), with removal of reverse (decoy) proteins and potential contaminants. Intensities were $\log_2$ -transformed; null or negative SILAC ratios were treated as missing before the logarithmic transformation. For the directDIA matrix, undetected values (labeled *Filtered*) were converted to missing and, when required for analyses needing a complete matrix, imputed by a Perseus-style approach (replacement by the lower tail of each sample's distribution, at $mean - 1.8 \times standard\ deviation$).

3.2. Biological anchoring of the SILAC signal

Since the orientation of SILAC ratios can be ambiguous, the signal was anchored to known biology: inhibition of the MAPK module must **reduce** phosphorylation of the pathway's canonical sites. An anchor set of approximately 30 MAPK/ERK pathway genes and their canonical substrates was defined; the sign convention was fixed so that the median of the anchor sites is negative (suppression). This check simultaneously provides a quality control independent of the processing.

3.3. Kinase-Substrate Enrichment Analysis (KSEA)

The activity of each kinase was estimated by KSEA: for a kinase k with a set of measured substrates S_k , the score is the standardized deviation of the substrate mean relative to the global mean, in the form $z = (m_S - m_{global}) / (\sigma_{global} / \sqrt{n})$. Significance was assessed by a permutation test (1,000 resamplings of substrate sets of the same size), and only kinases with at least five measured substrates were scored. The substrate sets come from the OmniPath network.

3.4. Differential analysis and statistics

Comparisons between groups (e.g. BRAF V600E vs. NRAS) were performed by Welch's t-test site by site, with effect magnitude expressed as the difference of means in $\log_2$ ($\log_2FC$). Sites were considered differentially phosphorylated with $p < 0.05$ and $|\log_2FC| > 1$. For the time course, the monotonicity of the response was measured by Spearman correlation between treatment time and expression, with Bonferroni correction; resistance signatures were defined as genes with $|p| > 0.7$ and adjusted $p < 0.01$ concordantly in both cell lines.

3.5. Dimensionality reduction and stratification

Principal Component Analysis (PCA) was applied, after standardization, to visualize the sample structure — separation between cell lines and the resistance trajectory over time. Molecular stratification of the TCGA-SKCM cohort followed the priority *BRAF* V600 > *NRAS* > *NF1* > triple wild-type, and overall survival was estimated by the Kaplan-Meier method per subtype.

3.6. Predictive modeling by neural network

A classifier was built to predict whether a phosphorylation site is differentially phosphorylated between the *BRAF* V600E and *NRAS* contexts, from the intensities across the six cell lines. The set was balanced (equal number of differential and non-differential sites) and split into training and test in a stratified manner (70/30); standardization was fit only on the training set to avoid information leakage. The architecture is a multilayer perceptron (6 → 128 → 64 → 32 → 1) with batch normalization, ReLU activation and 0.3 *dropout*; training used binary cross-entropy loss, the AdamW optimizer with weight decay and a cosine-annealing learning-rate scheduler, over 120 epochs. The training and test loss and ROC-AUC curves were recorded per epoch.

4. Phase 1 — Acute Suppression (6–24 h)

Treatment window: 6 h – 24 h | **Hallmark:** MAPK pathway shutdown; KSEA scores and ERK1/2 activation drop; the cell is driven into full target blockade.

The first phase of combined BRAFi + MEKi treatment is the immediate, on-target collapse of MAPK/ERK signaling. Evidence comes from the acute (30-minute) inhibitor phosphoproteome of A375 cells (PXD013923) and from Kinase-Substrate Enrichment Analysis (KSEA) over the OmniPath network, which together confirm that the drugs hit the intended module and shut it down.

Data window. The available acute phosphoproteome measures a 30-minute treatment, which lies just before the diagram's 6–24 h window; it captures the on-target biochemical shutdown that initiates the acute-suppression phase. No 6–24 h phosphoproteome exists for this system, so this earliest available time point is used as the Phase-1 readout.

Key result. Canonical MAPK/ERK phosphosites are selectively suppressed (anchor median $\log_2 \approx -0.29$ to -0.31 across the three inhibitors) while the rest of the phosphoproteome stays near zero, and KSEA scores the core kinases as strongly negative — the signature of full acute pathway shutdown.

4.1. Acute phosphoproteome response (PXD013923)

After quality filtering, 10,273 phosphorylation sites were quantified; for each inhibitor the mean change in phosphorylation ($\log_2$) relative to control was computed. The distribution is centered near zero but has a pronounced negative tail: 346 (BRAFi), 331 (MEKi) and 290 (ERKi) sites are strongly suppressed ($\log_2 < -1$), against far fewer induced sites.

Acute phosphoproteome response to MAPK-pathway inhibition (A375, 30 min)

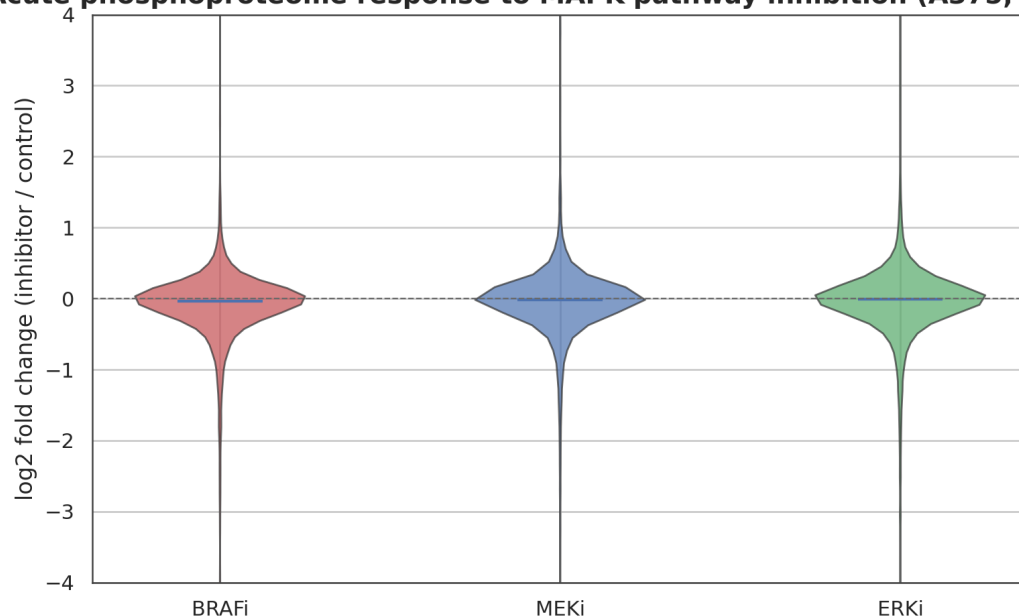


Figure 4.1. Distribution of phosphorylation changes ($\log_2$, inhibitor/control) for BRAFi, MEKi and ERKi in A375 (30 min). The negative tail concentrates the suppressed sites.

4.2. Linearity of the RAF→MEK→ERK module

The responses to the three inhibitors are strongly correlated: Pearson r of 0.756 (BRAFi vs. MEKi), 0.705 (BRAFi vs. ERKi) and 0.760 (MEKi vs. ERKi). This is the expected signature of a linear module, in which inhibition at any level propagates similar effects downstream.

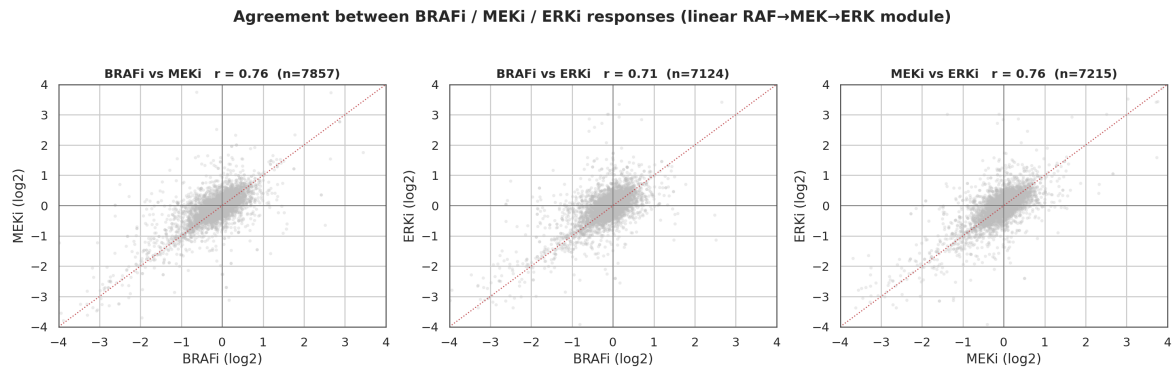


Figure 4.2. Pairwise agreement between the responses to BRAFi, MEKi and ERKi. The strong positive correlation reflects the linearity of the RAF→MEK→ERK module.

4.3. Selective suppression of the pathway (anchor)

Against the rest of the phosphoproteome, the MAPK/ERK pathway sites shift systematically negative across all three inhibitors (median of -0.293 for BRAFi, -0.267 for MEKi and -0.310 for ERKi; anchor set of 81 sites, overall median -0.295), while the remaining sites stay centered at zero — the suppression is specific to the target pathway. Among the most repressed BRAFi targets are the ERK-regulated repressor ERF and the ERK effector RPS6KA3 (RSK2).

Canonical MAPK/ERK phosphosites are selectively suppressed by inhibition

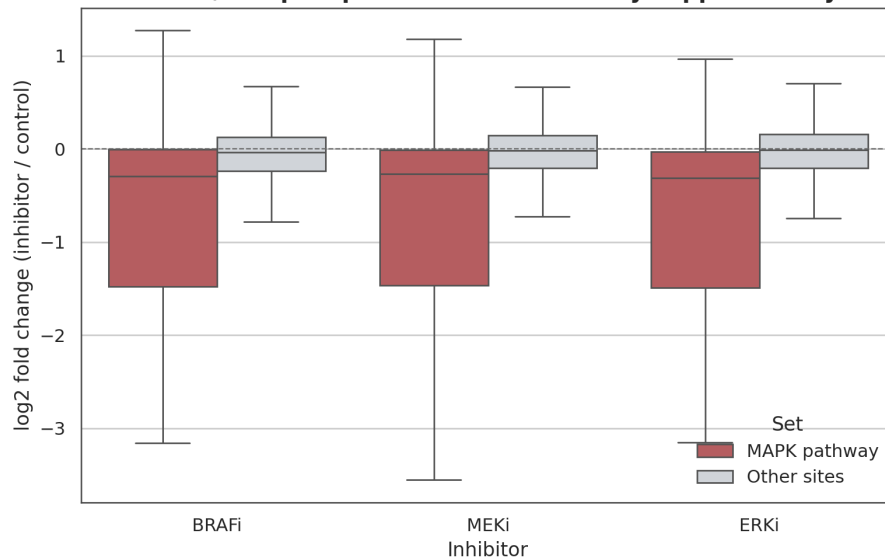


Figure 4.3. MAPK/ERK pathway sites (red) vs. other sites (grey). The suppression concentrates in the pathway's canonical sites.

Site (gene_residue)	Gene	log ₂ FC (BRAFi)
CCDC175_T247	CCDC175	-6.41
ENSP00000330271_Y137	ENSP00000330271	-5.93

Site (gene_residue)	Gene	log ₂ FC (BRAFi)
ERF_T7	ERF	-5.62
CHST13_S312	CHST13	-5.43
GIGYF2_S242	GIGYF2	-4.99
TLR10_S436	TLR10	-4.31
RAI14_S264	RAI14	-4.16
ENSP00000367263:map.11/6...	ENSP00000367263:map....	-4.15

Table 4.1. Phosphorylation sites most suppressed by BRAFi (selection).

4.4. Kinase-activity collapse (KSEA)

KSEA over the OmniPath kinase-substrate network translates the measured sites into per-kinase activity scores. Under BRAFi, the core of the pathway shows the most negative scores — BRAF ($z = -6.6$), RAF1 ($z = -6.5$), MAP2K1/MEK1 ($z = -6.5$) and MAPK1/3 — all with $p < 0.001$ in the permutation test, reconstructing the pathway's suppression hierarchy. Negative regulators such as the DUSP phosphatases (DUSP1/8/16) and the scaffold KSR1, whose phosphorylation is ERK-dependent, drop the most sharply.

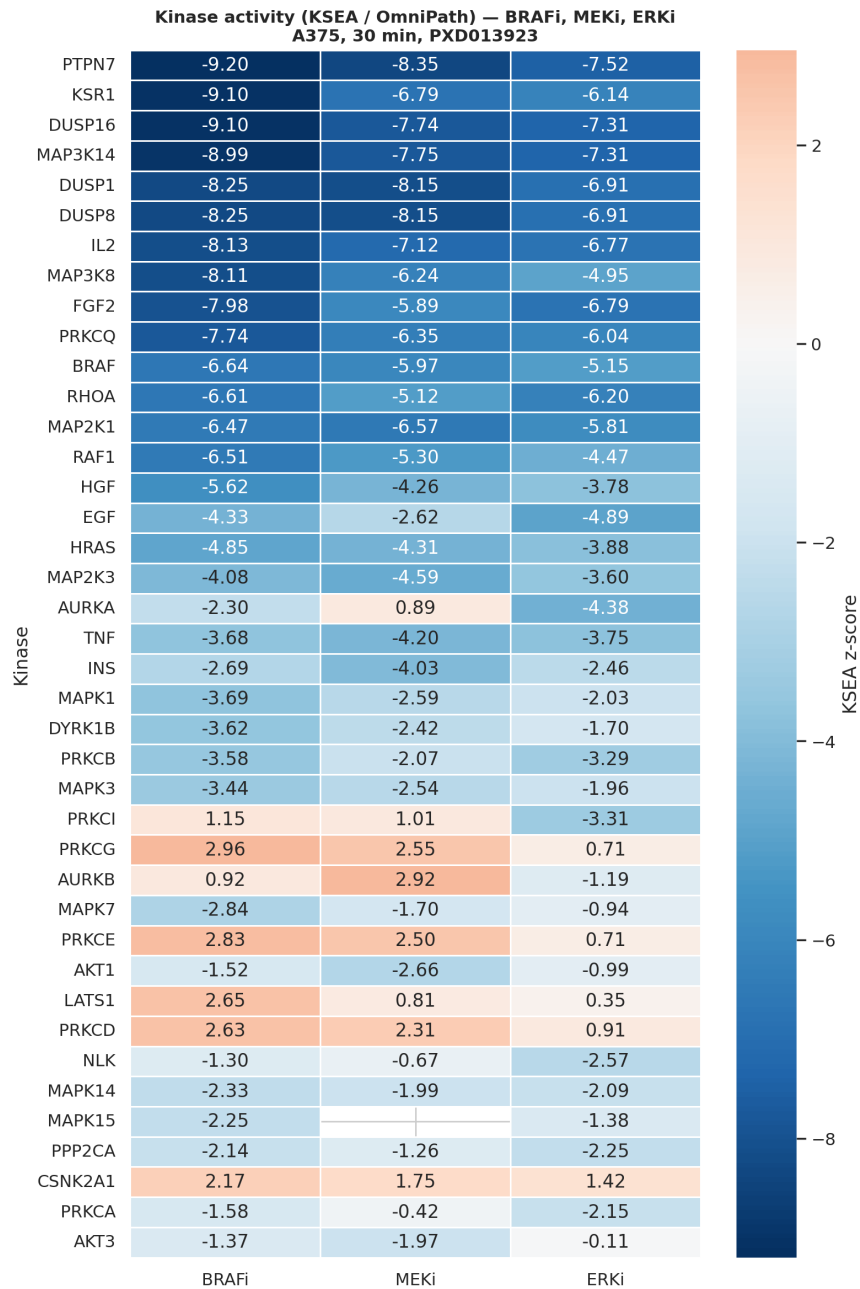


Figure 4.4. KSEA scores (z) per kinase under BRAFi, MEKi and ERKi. Blue tones indicate suppression of kinase activity.

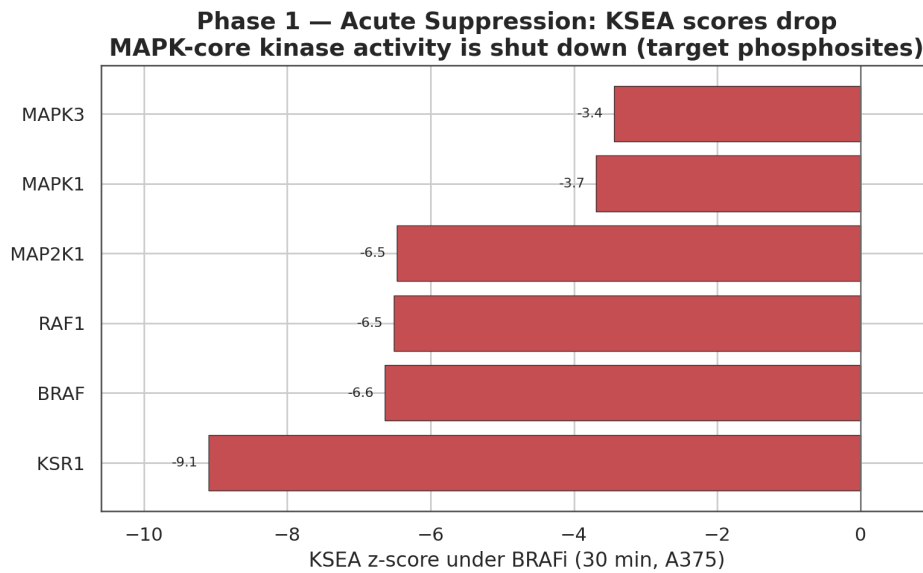


Figure 4.5. Phase-1 signature: KSEA scores of the MAPK core drop sharply under BRAFi — the quantitative form of the 'KSEA scores drop / MAPK pathway shutdown' panel of the model.

Kinase	z (BRAFi)	z (MEKi)	z (ERKi)
PTPN7	-9.20	-8.35	-7.52
KSR1	-9.10	-6.79	-6.14
DUSP16	-9.10	-7.74	-7.31
MAP3K14	-8.99	-7.75	-7.31
DUSP1	-8.25	-8.15	-6.91
DUSP8	-8.25	-8.15	-6.91
IL2	-8.13	-7.12	-6.77
MAP3K8	-8.11	-6.24	-4.95
FGF2	-7.98	-5.89	-6.79
PRKCQ	-7.74	-6.35	-6.04

Table 4.2. Ten most-suppressed kinases under BRAFi (KSEA z-score).

5. Phase 2 — Adaptive Rewiring (48-72 h)

Treatment window: 48 h – 72 h | **Hallmark:** Loss of negative feedback triggers emergency activation of parallel routes; PI3K/AKT/mTOR and RTK (AXL, PDGFR) activities rebound; the cell begins to bypass the block.

Once the acute block is established, the cell escapes it. The loss of ERK-dependent negative feedback releases upstream receptors, and alternative kinase routes are activated to sustain survival and proliferation. This phase gathers three complementary readouts of that rewiring: kinases that rebound after acute suppression, the reactivation of receptor tyrosine kinases, and the parallel-pathway kinase activity that characterizes MAPK-independent signaling states.

Key result. Several kinases that are acutely suppressed by inhibition (KSR1, MAP3K14, DUSP8, DYRK1B, PRKCA) have their expression reinforced in the resistant state, and receptor tyrosine kinases (AXL, PDGFRB, MET, EGFR) are progressively reactivated — the molecular signature of emergency activation of parallel routes.

Data window. The rewiring window (48-72 h) has no dedicated dataset; its molecular signature is reconstructed from the earliest arms of the resistance time course (GSE110054) and from the cross-genotype baseline (PXD022992). The pathway-score trajectories below (Figure 5.3) show the rise of parallel routes beginning within the first days of treatment.

5.1. Rebounding kinase activities

Cross-referencing the acute kinase suppression (KSEA at 30 min, PXD013923) with the late transcriptional change (expression in the resistant state, GSE110054) isolates kinases that, although suppressed immediately by the drug, have their gene expression reinforced once resistance sets in — natural candidates for the adaptive escape that drives rewiring. The most prominent are KSR1, MAP3K14, DUSP8, DYRK1B and PRKCA.

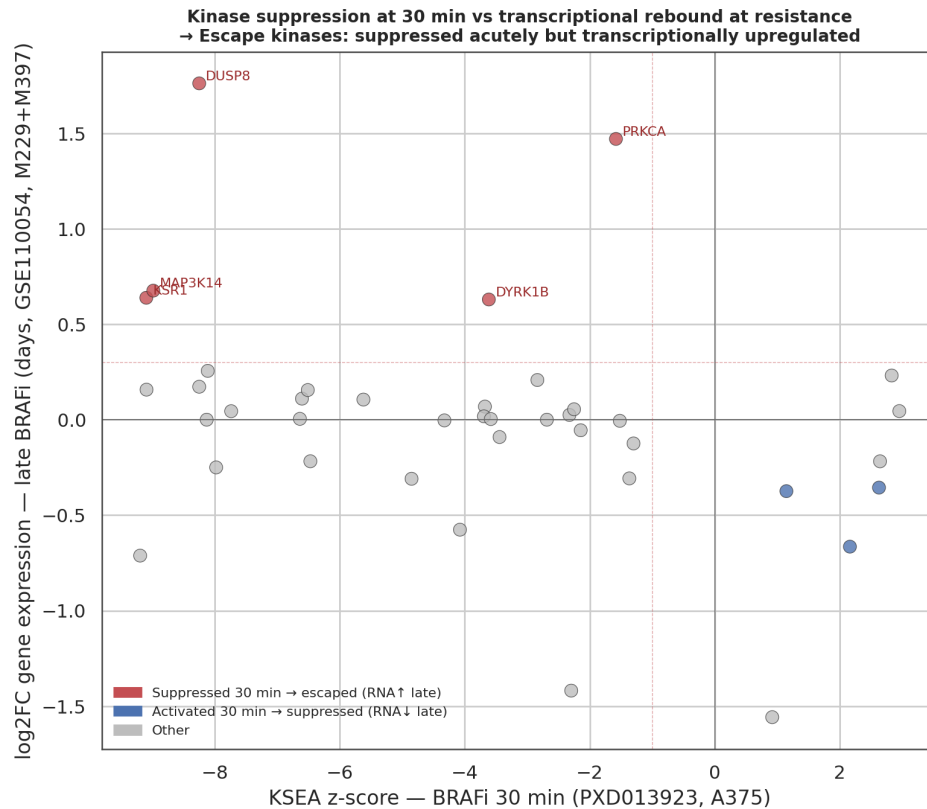


Figure 5.1. Acute kinase suppression (x-axis, KSEA 30 min) vs. late transcriptional rebound (y-axis, resistant state). Escape kinases — suppressed acutely but re-expressed late — are highlighted.

5.2. Receptor tyrosine-kinase reactivation

The hallmark of adaptive rewiring is the reactivation of receptor tyrosine kinases (RTKs) that feed the parallel PI3K/AKT and RAS branches. Along the treatment time course, RTK genes — including AXL, PDGFRB, MET, EGFR and FGFR1 — rise progressively, together with the compensatory increase of AKT/mTOR-axis members, consistent with the emergency activation of survival routes as the MAPK block is bypassed.

Key resistance gene trajectories — M229 and M397 under BRAFi

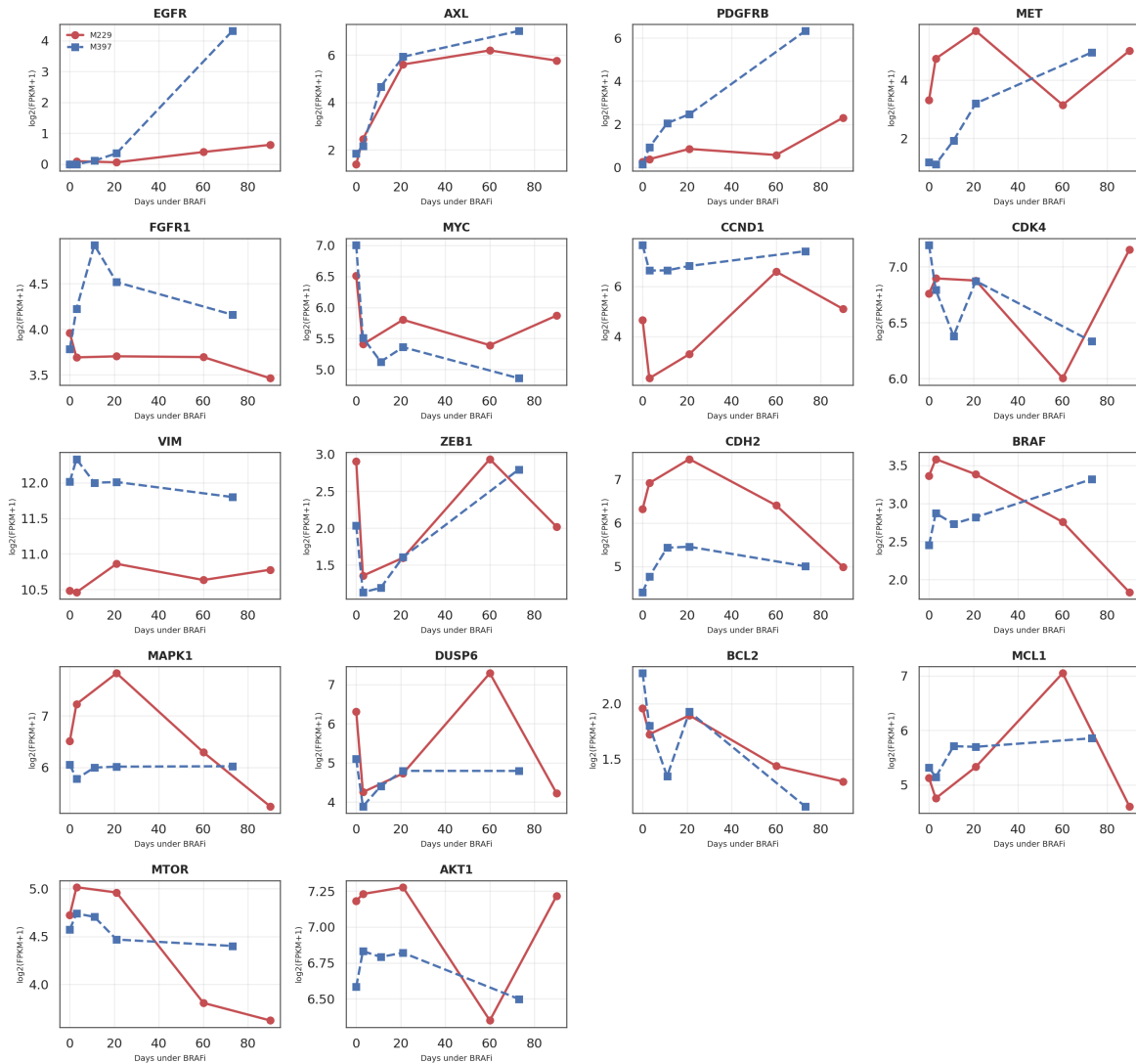


Figure 5.2. Temporal trajectories of key resistance genes in M229 and M397 under BRAFi — receptor tyrosine kinases (AXL, PDGFRB, MET, EGFR, FGFR1), proliferation, EMT and survival markers rise as the cell rewires.

Aggregating these genes into pathway scores (mean change vs. the DMSO baseline) reproduces the model's rising panels directly: the PI3K/AKT/mTOR and RTK scores climb above baseline within days, while MAPK output — sharply suppressed at the first treated time point — rebounds later, the transcriptional counterpart of the phosphoproteomic rebound. EMT/invasion scores rise in parallel.

Signaling-transition kinetics — pathway activity over the BRAFi time course
MAPK output drops then rebounds; PI3K/AKT/mTOR, RTK and EMT rise (GSE110054, M229 & M397)

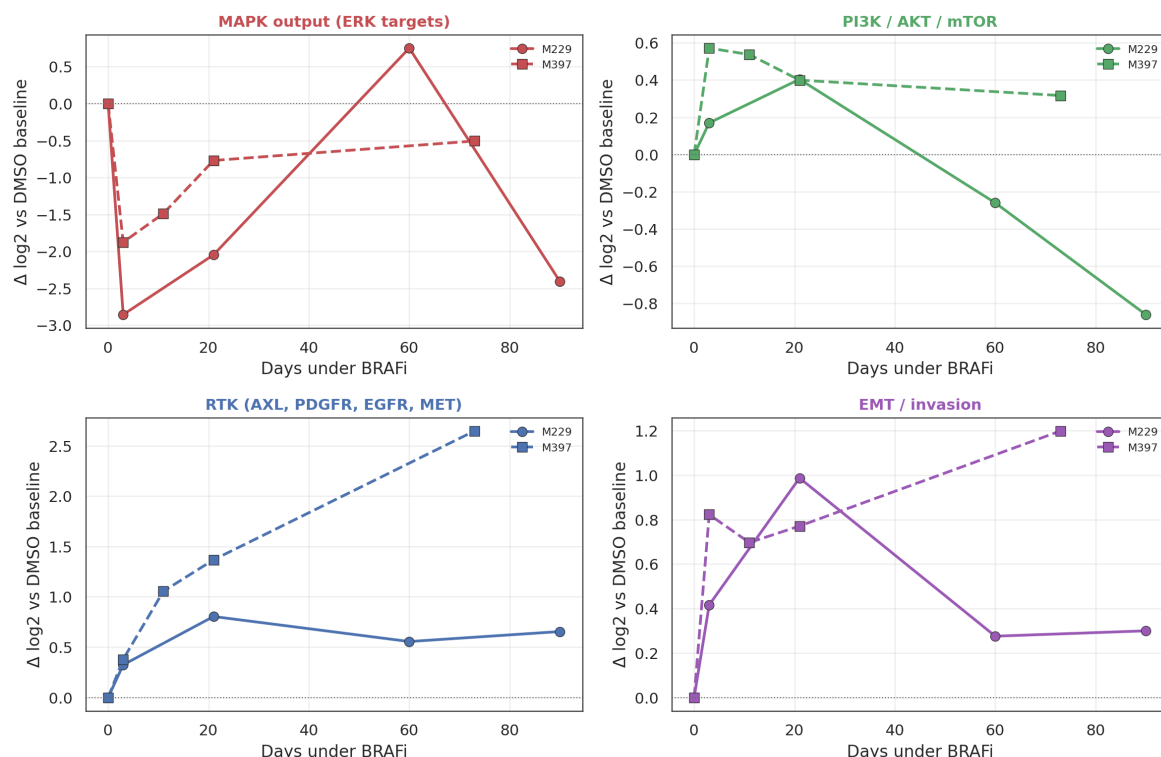


Figure 5.3. Pathway-activity trajectories over the BRAFi time course ($\Delta\log_2$ vs. DMSO, GSE110054): MAPK output drops then rebounds; PI3K/AKT/mTOR, RTK (AXL/PDGFR/EGFR/MET) and EMT/invasion rise — the diagram's rising panels realized in real data.

5.4. Parallel-pathway signaling across genotypes (PXD022992)

The directDIA phosphoproteome of six melanoma cell lines provides a static model of the rewired, MAPK-independent state: the NRAS-mutant lines survive without a druggable BRAF V600E node and therefore exemplify the parallel-route signaling that BRAFi-resistant cells acquire. Kinase-activity scores by substrate set (ERK, AKT, mTOR, CDK and FAK/SRC) summarize how signaling is distributed differently across the genotypes, with the PI3K/AKT/mTOR and FAK/SRC axes prominent in the non-BRAF-V600E context.

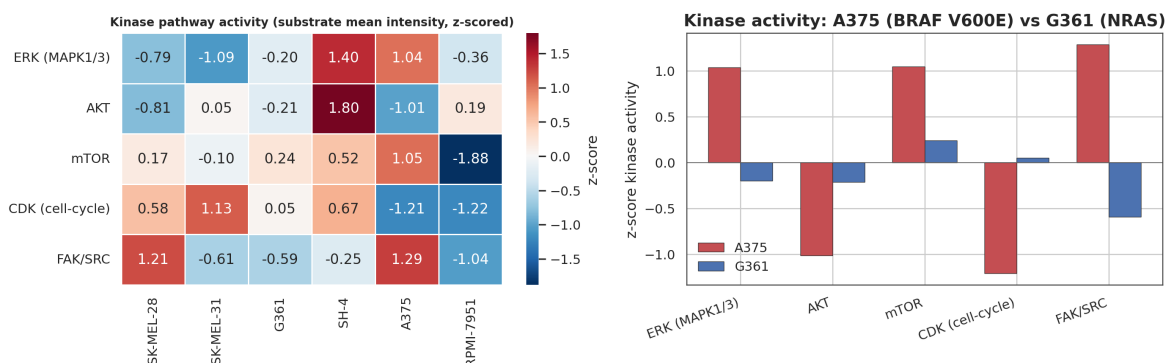


Figure 5.4. Kinase-pathway activity per cell line (substrate mean, standardized): ERK, AKT, mTOR, CDK and FAK/SRC.

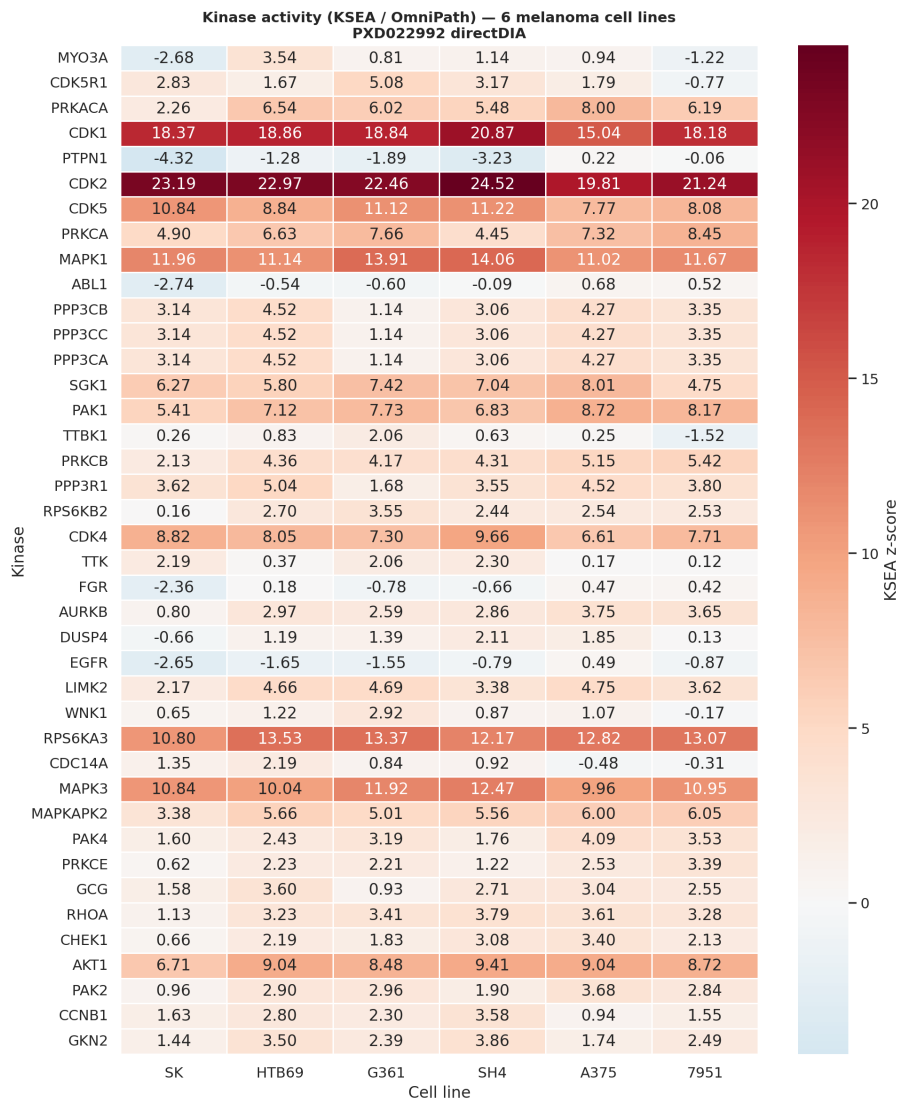


Figure 5.5. KSEA scores per kinase across the six melanoma cell lines.

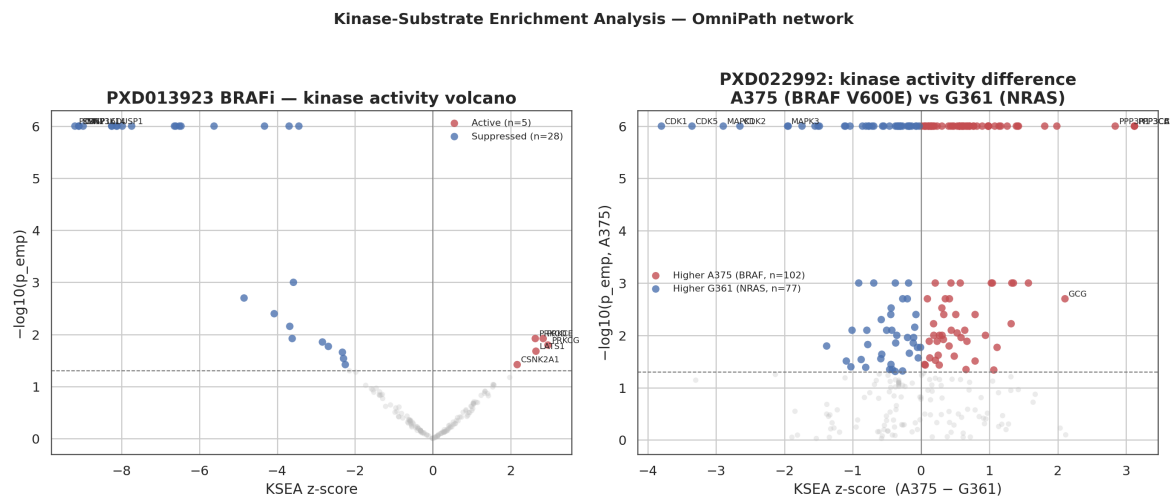


Figure 5.6. Comparative panel: kinase activity under BRAFi (left) and the difference in kinase activity between A375 (BRAF V600E) and G361 (NRAS) (right).

6. Phase 3 — Consolidated Resistance (3-90 days)

Treatment window: 3 d – 90 d | **Hallmark:** Stable, fully integrated alternative signaling; concordant multi-omics signatures; phosphoproteomic rebound and phenotypic switching toward invasive cells.

In the final phase the rewired state is consolidated into a stable, heritable resistant phenotype. The evidence spans the long-term transcriptional time course (GSE110054), the stable genotype-specific phosphoproteome (PXD022992), the clinical tumor landscape (TCGA-SKCM), and a predictive model that shows the resistant signature is learnable.

Key result. The resistant state carries a robust, concordant signature — 647 genes consistently induced and 267 repressed across two cell lines — with induction of EMT/invasion programs (phenotypic switching). The stable genotype-specific phosphoproteome is strong enough that a neural network predicts differential sites at test ROC-AUC 0.989.

Data window. This is the one phase whose data window matches the model exactly: the GSE110054 time course spans 3 days to 90 days. The MAPK-output rebound seen in Figure 5.3 — sharp suppression at the first treated point followed by recovery toward or above baseline — is the transcriptional readout of the model's 'phosphoproteomic rebound'. A direct phospho time course is not available for this system.

6.1. Trajectory to the resistant state (GSE110054)

The time course in M229 and M397 under vemurafenib (DMSO → 3 d → 11–21 d → 73–90 d) traces the progressive transcriptional reprogramming of resistance. Principal-component analysis organizes the samples along a continuous trajectory from the sensitive state (DMSO) to acquired resistance, and the quality control confirms consistent, well-correlated samples.

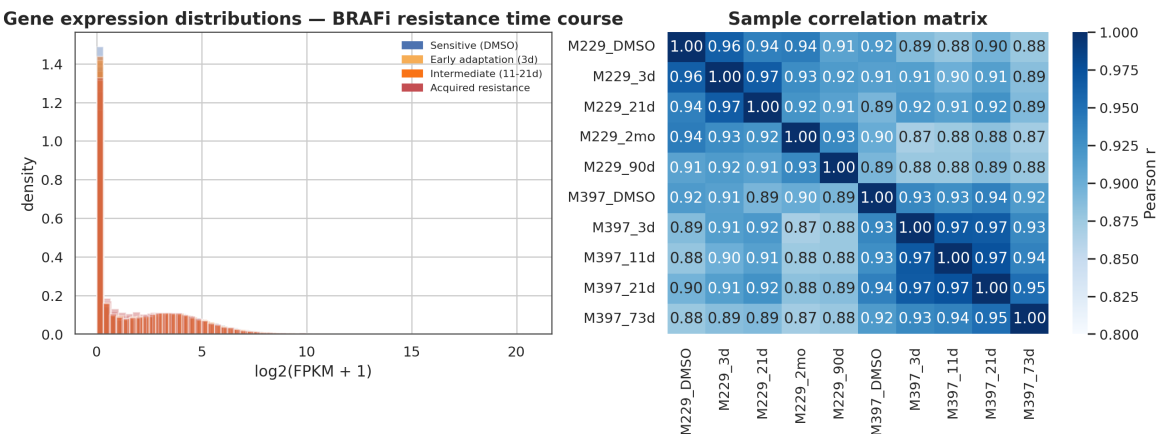


Figure 6.1. Quality control of the time course: expression distributions and sample correlation matrix.

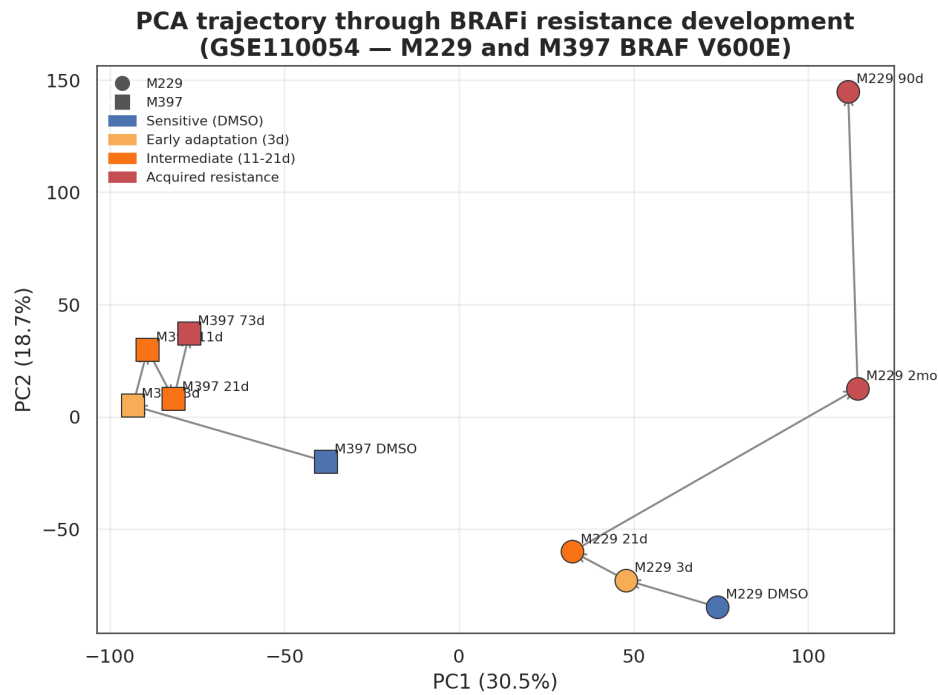


Figure 6.2. PCA trajectory from sensitivity to resistance in M229 and M397 under BRAFi. The arrows indicate temporal progression.

6.2. Stable, concordant signature and phenotypic switch

A monotonicity analysis (Spearman) identifies, concordantly in both cell lines, 647 genes consistently induced and 267 consistently repressed over treatment ($|\rho| > 0.7$). The pathway dynamics show the reactivation of RTK and PI3K/AKT/mTOR programs and the induction of EMT/invasion markers — the transcriptional face of the phenotypic switch toward invasive cells that defines consolidated resistance.

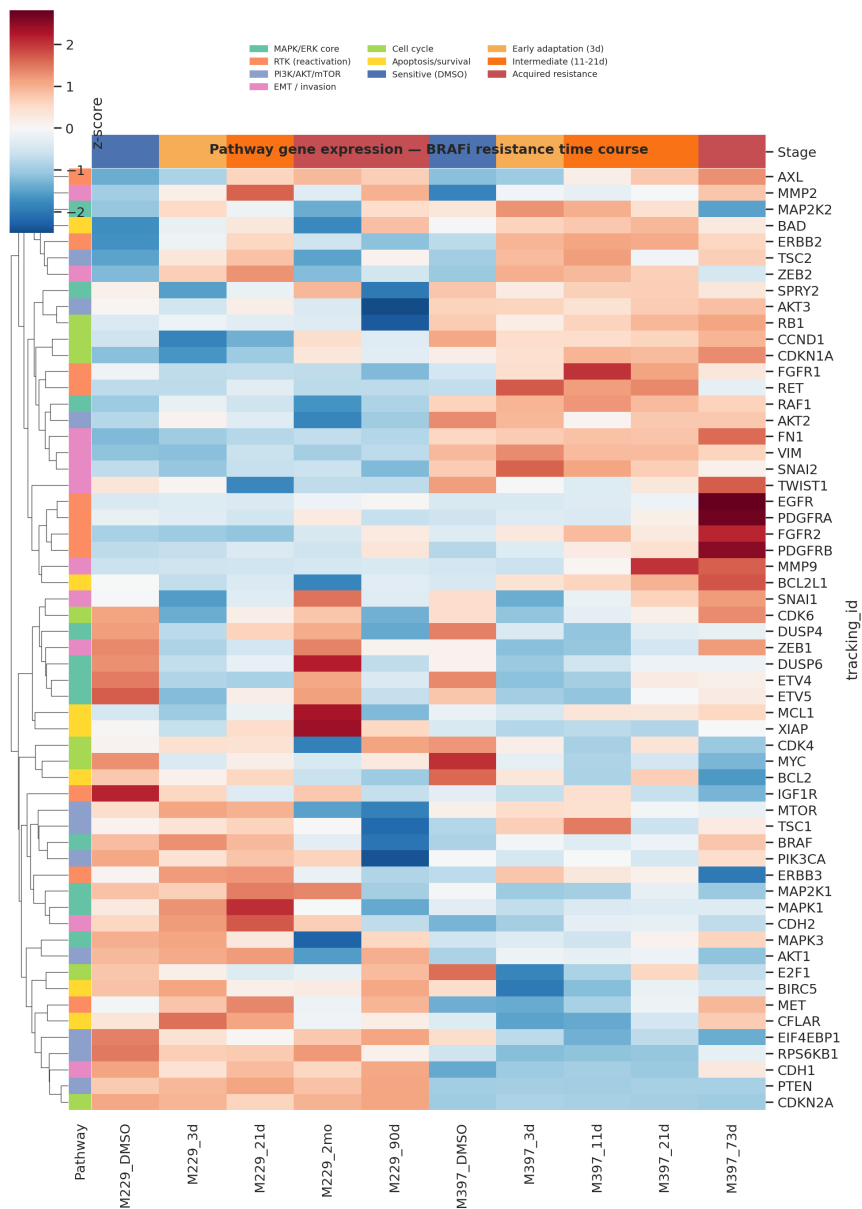


Figure 6.3. Pathway expression dynamics over the BRAFi time course.

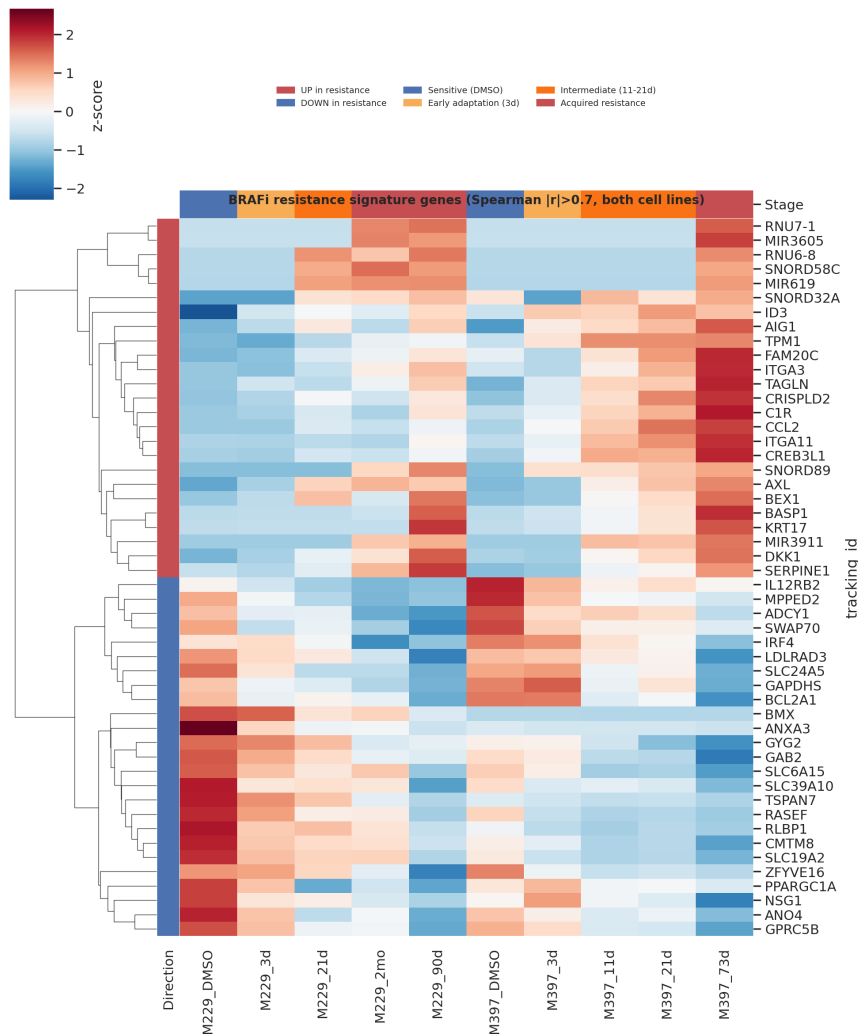


Figure 6.4. Resistance signature: genes consistently induced and repressed ($|\rho| > 0.7$ in both cell lines).

6.3. Consolidated genotype-specific phosphoproteome (PXD022992)

The basal phosphoproteome of the six cell lines (55,939 sites; intensity medians homogeneous between 14.59 and 14.74) captures the stable signaling state of each genotype. The comparison between BRAF V600E and NRAS-mutant lines identifies 1,387 sites with higher phosphorylation in BRAF V600E and 1,304 higher in NRAS (Welch's t-test; $p < 0.05$ and $|\log_2 FC| > 1$), and PCA separates the two contexts along PC1 — a reproducible, stable phosphorylation signature of the resistant genotype.

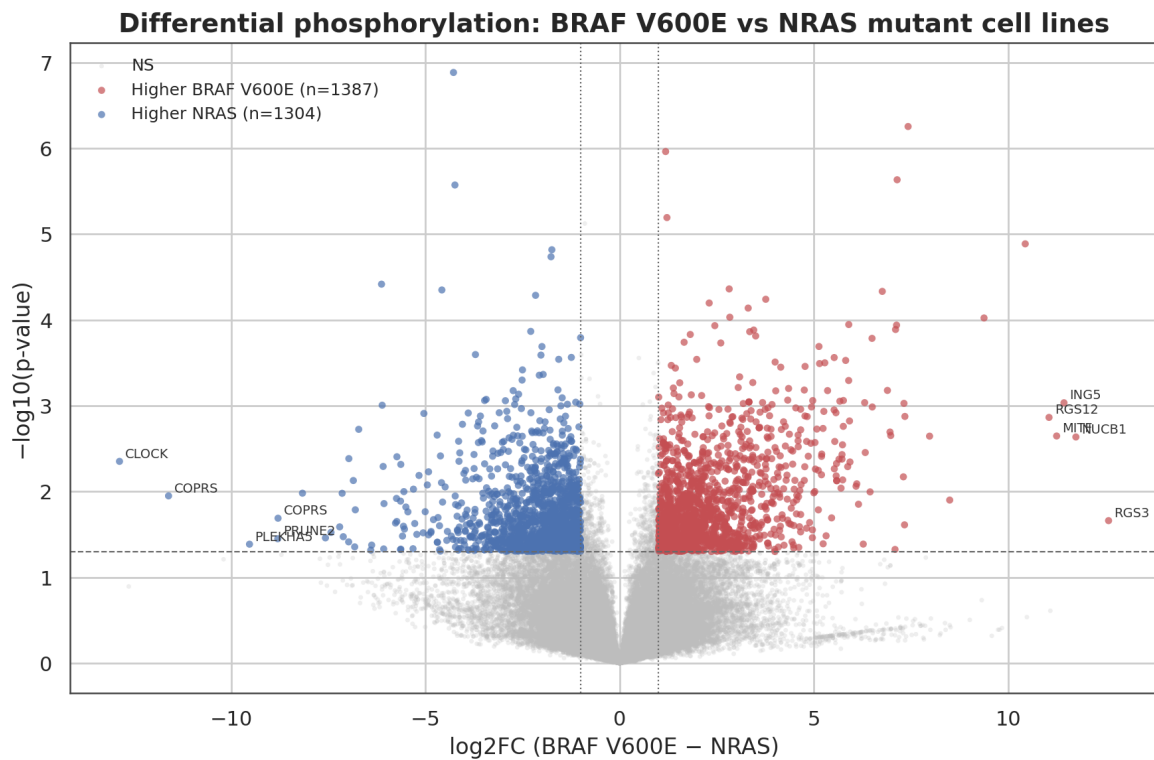


Figure 6.5. Volcano plot of differential phosphorylation between BRAF V600E and NRAS-mutant cell lines.

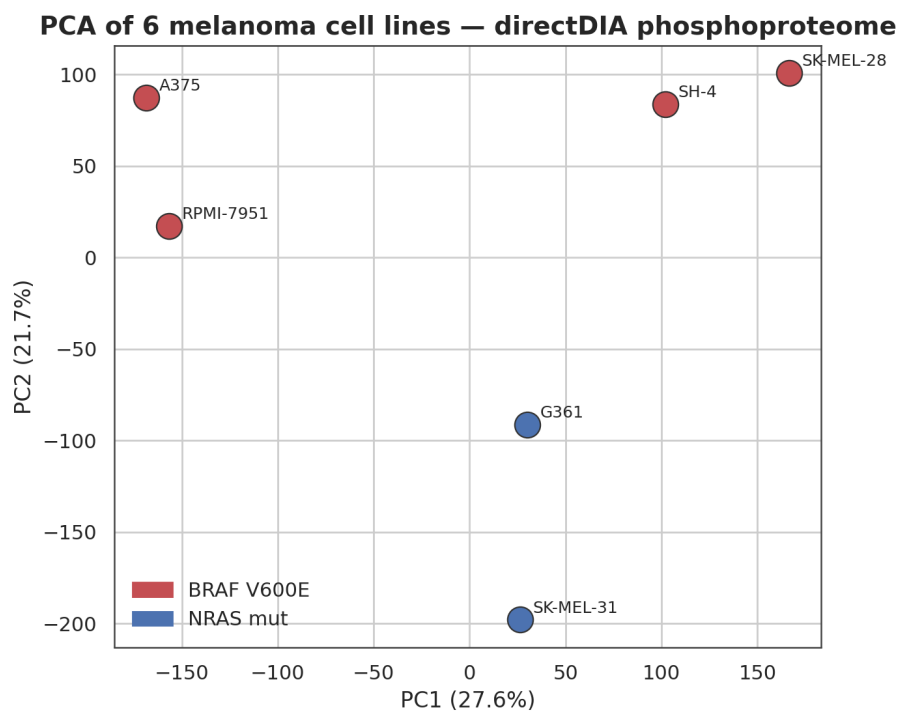


Figure 6.6. PCA of the six melanoma cell lines; PC1 follows the mutational context (BRAF V600E vs. NRAS).

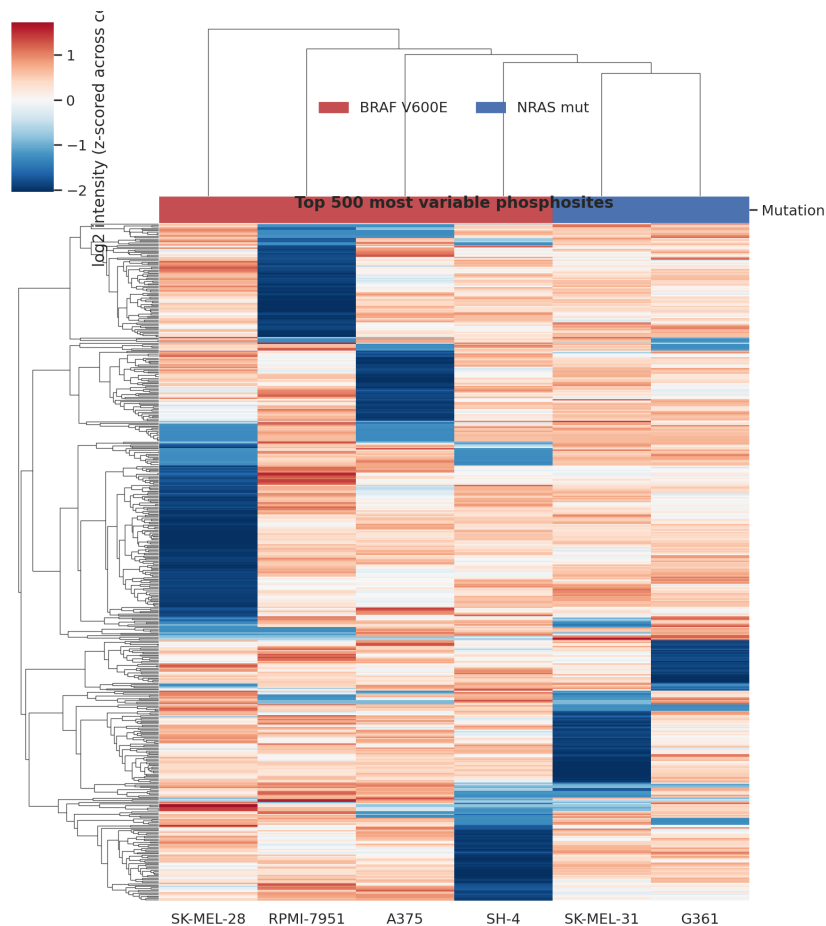


Figure 6.7. Heatmap of the highest-variance phosphorylation sites, with hierarchical clustering of the cell lines.

6.4. Clinical anchoring (TCGA-SKCM)

The TCGA-SKCM cohort (470 patients) anchors the consolidated resistant state in human tumors. The driver-gene mutation frequencies recover the canonical melanoma pattern — *BRAF* most frequent, followed by *NRAS*, with *NF1*, *CDKN2A* and *PTEN* — and the molecular subtypes organize the disease's heterogeneity.

Gene	Mutated patients	Frequency (%)
BRAF	235	50.0
NRAS	127	27.0
PREX2	96	20.4
NF1	63	13.4
CDKN2A	57	12.1
PTEN	44	9.4
PPP6C	30	6.4
RAC1	29	6.2

Table 6.1. Mutation frequency of the main driver genes in TCGA-SKCM.

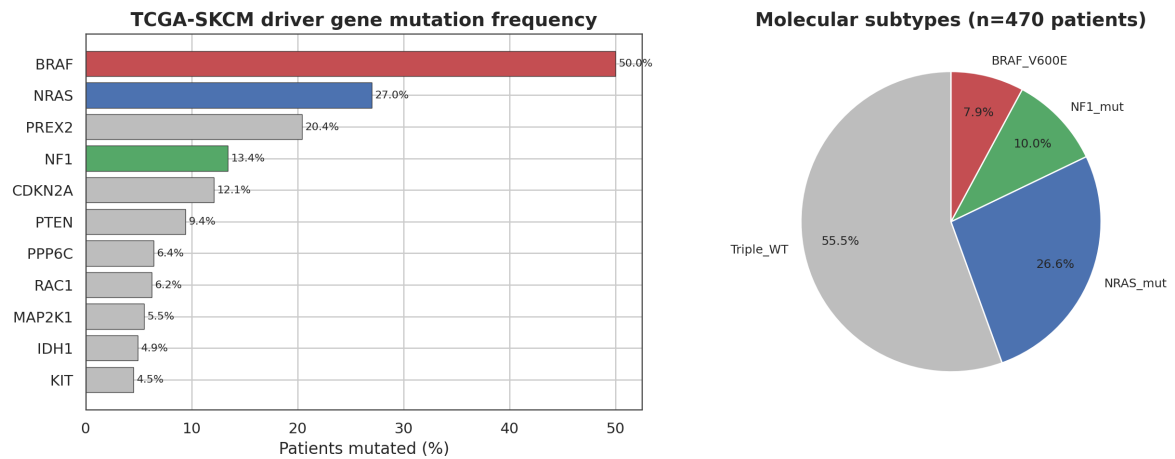


Figure 6.8. Mutational landscape of TCGA-SKCM: driver-gene frequency (left) and molecular-subtype distribution (right).

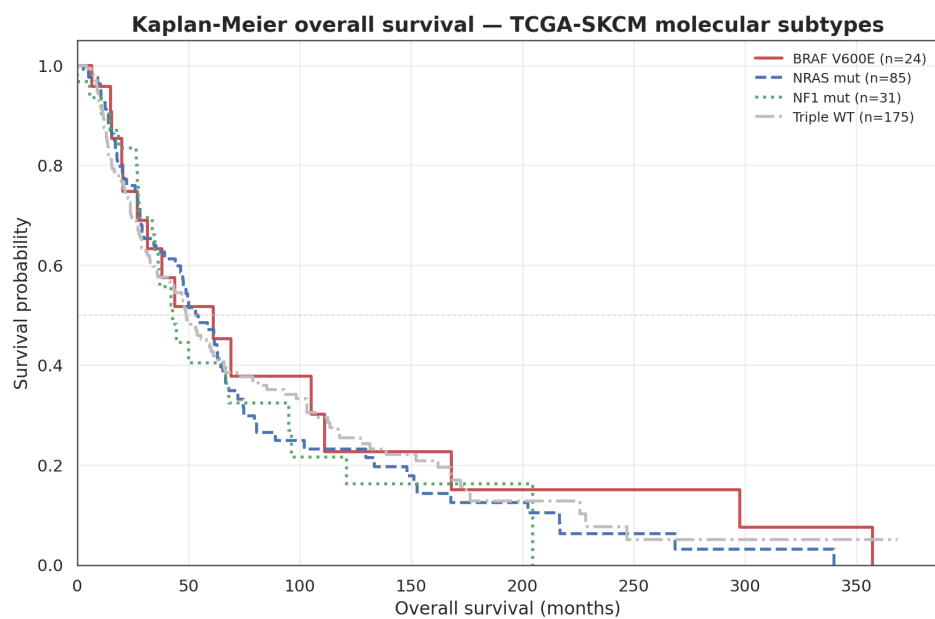


Figure 6.9. Overall survival (Kaplan-Meier) by molecular subtype in TCGA-SKCM.

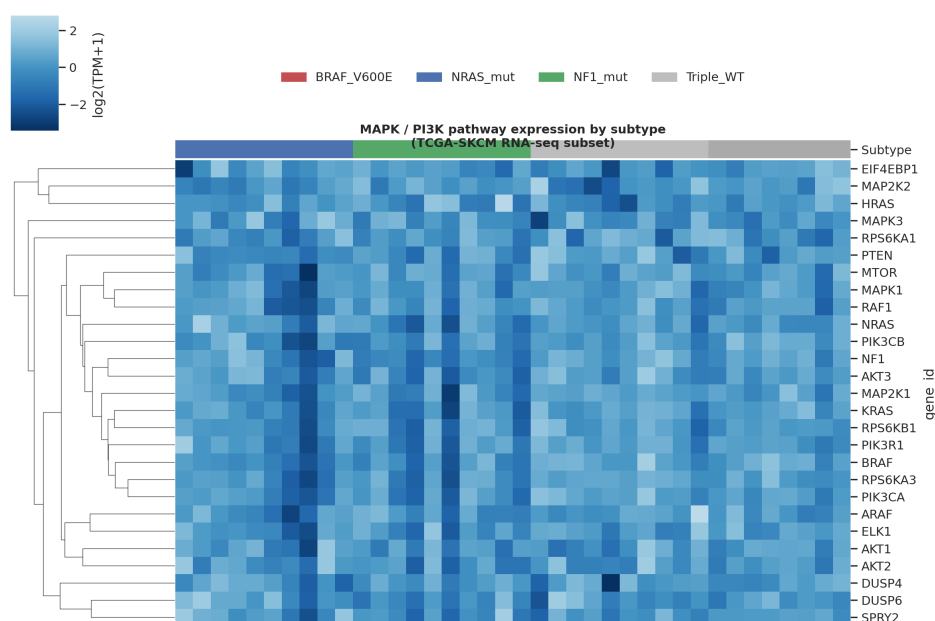


Figure 6.10. Expression of MAPK/PI3K pathway genes by molecular subtype (RNA-seq, TCGA-SKCM).

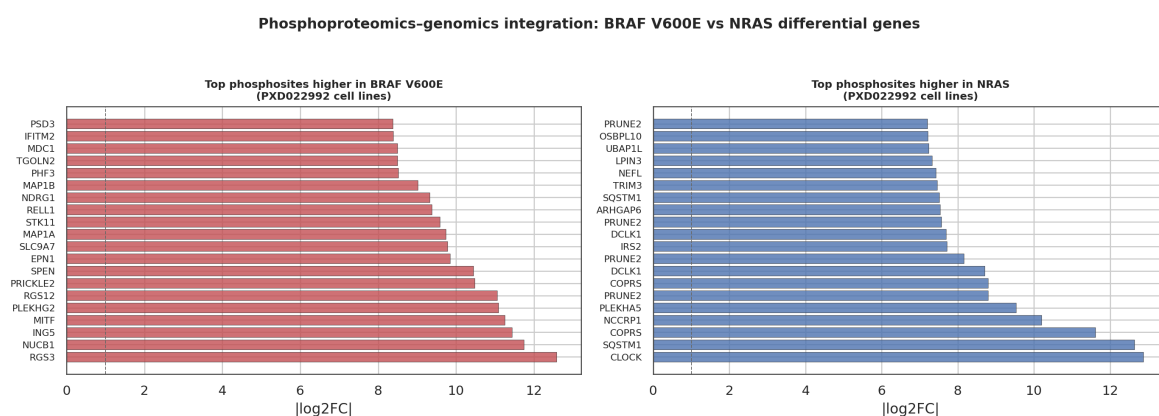


Figure 6.11. Phosphoproteomics-genomics integration: main differential genes between BRAF V600E and NRAS.

6.5. Predictive model of the resistant signature

To test whether the stable phosphorylation signature distinguishing BRAF V600E from NRAS is learnable, a neural network was trained on 5,382 balanced sites (stratified 3,767 train / 1,615 test; 11,649 parameters; 120 epochs). It reaches a test ROC-AUC of 0.989 (best epoch 0.991), F1 of 0.95 and 95% accuracy, with training and test curves that evolve closely — the resistant signature is not only stable but predictable.

Neural network — predict differentially phosphorylated sites
(PXD022992 DIA, BRAF V600E vs NRAS, 6 melanoma cell lines)

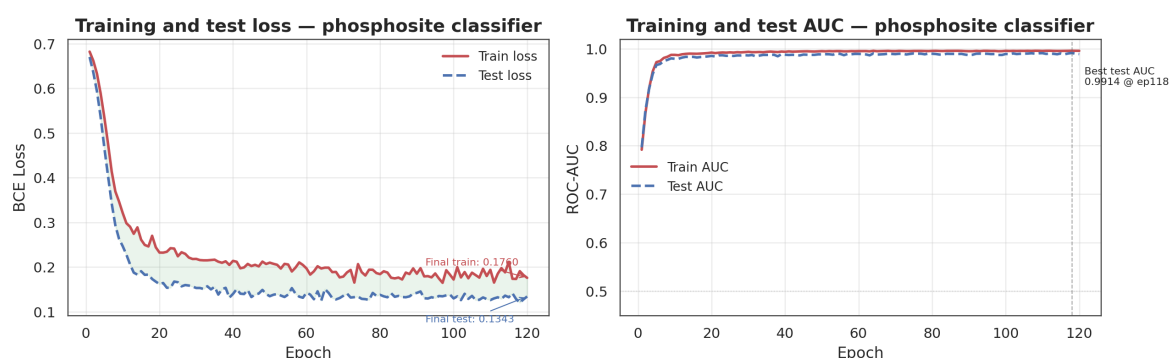


Figure 6.12. Neural-network training and test curves: loss (left) and ROC-AUC (right) per epoch. The closeness of the curves indicates good generalization.

Test set evaluation — ROC-AUC 0.9891 | Average Precision 0.9851

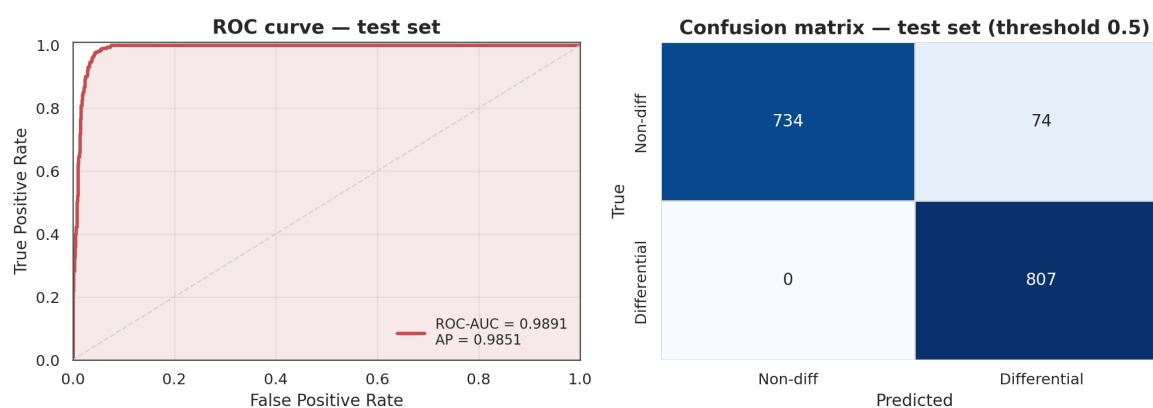


Figure 6.13. Test evaluation: ROC curve (left) and confusion matrix at the 0.5 threshold (right).

Epoch	Train loss	Test loss	Train AUC	Test AUC
1	0.6821	0.6701	0.7917	0.7965
10	0.3211	0.2468	0.9874	0.9794
30	0.2183	0.1533	0.9937	0.9863
60	0.1969	0.1428	0.9952	0.9894
90	0.1852	0.1423	0.9950	0.9869
120	0.1760	0.1343	0.9958	0.9891

Table 6.2. Evolution of loss and ROC-AUC on training and test over the course of training.

7. Multi-omics integration and concluding remarks

Taken together, the analyses trace a single coherent arc through the three phases of the signaling-transition-kinetics model, connecting the immediate drug effect to the stable resistant phenotype:

- **Phase 1 — Acute Suppression (6-24 h).** The acute phosphoproteomics (PXD013923) and KSEA demonstrate, with statistical robustness, the selective and linear suppression of the RAF→MEK→ERK module by inhibition, validating the entire processing workflow on purely biological grounds.
- **Phase 2 — Adaptive Rewiring (48-72 h).** Kinases suppressed acutely rebound transcriptionally (KSR1, MAP3K14, DUSP8, DYRK1B, PRKCA), receptor tyrosine kinases (AXL, PDGFRB, MET) are reactivated, and cross-genotype kinase activity exposes the PI3K/AKT/mTOR and FAK/SRC parallel routes — the emergency activation that bypasses the block.
- **Phase 3 — Consolidated Resistance (3-90 d).** The time course (GSE110054) yields a stable, concordant signature (647 up, 267 down) with EMT/invasion induction (phenotypic switching); the stable genotype-specific phosphoproteome (PXD022992) and the clinical TCGA-SKCM cohort anchor the resistant state in reproducible and human tumor contexts.
- **Predictive modeling.** The neural network confirms that the consolidated phosphorylation signature is learnable and highly predictive (test ROC-AUC 0.989).

7.1. Linking acute suppression and late adaptation

One of the most informative results emerges from the integration between acute kinase suppression and late transcriptional rebound: kinases such as KSR1, MAP3K14, DUSP8, DYRK1B and PRKCA, although suppressed immediately by inhibition, have their expression reinforced in the resistant state. This pattern links, along a single analytical axis, the minutes-scale biochemical response to the months-scale transcriptional adaptation, and points to natural candidates for the investigation of escape mechanisms.

7.2. Reproducibility

All steps are reproducible: the data come from established public repositories (PRIDE, GEO and GDC/TCGA); the processing, analyses and generation of figures and tables are organized into independent modules, orchestrated by a central script with logging and standardized saving of results. Auxiliary mappings (such as identifier conversion and the kinase-substrate network) are stored in a local cache, ensuring deterministic execution.

7.3. Synthesis

The pipeline successfully integrates targeted and data-independent phosphoproteomics, temporal transcriptomics, clinical genomics and deep learning into a single, biologically consistent narrative about the response and resistance to BRAFi/MEKi in melanoma. The results faithfully recapitulate the known biology of the MAPK/ERK pathway, characterize the heterogeneity across mutational contexts, describe the temporal dynamics of resistance, and demonstrate the feasibility of predictive models from the observed molecular patterns, establishing a solid and extensible foundation for subsequent investigations.

8. Appendix — Inventory of figures and tables

All figures (PNG) and tables (CSV) referenced in this report were generated by the pipeline and are available in the *results/figures/* and *results/outputs/* directories. The main files are listed below.

8.1. Figures

File	Description
phase2_model_diagram.png	Signaling-transition-kinetics reference model
phase1_ksea_dropping_bars.png	Phase 1 — KSEA scores drop (MAPK core)
signaling_kinetics_pathway_trajectories.png	Pathway-activity trajectories over time
pxd013923_response_distributions.png	Acute response per inhibitor
pxd013923_inhibitor_agreement.png	Agreement between inhibitors
pxd013923_mapk_anchor.png	Anchoring in the MAPK/ERK pathway
pxd022992_qc.png	directDIA phosphoproteome QC
pxd022992_pca_cell_lines.png	PCA of cell lines
pxd022992_volcano_braf_vs_nras.png	Differential BRAF vs. NRAS
pxd022992_heatmap_top_variance.png	Highest-variance sites
pxd022992_ksea_kinase_activity.png	Kinase activity per cell line
ksea_pxd013923_inhibitors.png	KSEA under BRAFi/MEKi/ERKi
ksea_pxd022992_cell_lines.png	KSEA per cell line
ksea_combined_volcano.png	KSEA synthesis
tcga_skcm_mutation_landscape.png	TCGA mutational landscape
tcga_skcm_survival_km.png	Survival by subtype
tcga_skcm_pathway_expression.png	Pathway expression by subtype
tcga_skcm_phospho_integration.png	Phospho-genomics integration
temporal_gse110054_qc.png	Time-course QC
temporal_pca_resistance_trajectory.png	Resistance trajectory
temporal_pathway_heatmap.png	Pathway dynamics over time
temporal_key_gene_trajectories.png	Key gene trajectories
temporal_resistance_signature_heatmap.png	Resistance signature
temporal_kinase_escape_integration.png	Escape kinases
nn_training_curves.png	Neural-network training/test curves
nn_roc_confusion.png	ROC and confusion matrix

8.2. Result tables

File	Description
PXD013923_inhibitor_log2_matrix.csv	log ₂ matrix per inhibitor (10,273 sites)
PXD013923_top_suppressed_sites.csv	Most-suppressed sites
PXD022992_phosphosite_matrix.csv	Phosphorylation matrix (55,939 sites)
PXD022992_differential_phospho_BRAFvsNRAS.csv	Differential BRAF vs. NRAS
KSEA_PXD013923_kinase_zscores.csv	KSEA scores (inhibitors)
KSEA_PXD022992_kinase_zscores.csv	KSEA scores (cell lines)
signaling_kinetics_trajectory_scores.csv	Pathway-score trajectories over time
TCGA_SKCM_mutation_frequency.csv	Mutation frequency
TCGA_SKCM_clinical_with_subtypes.csv	Clinical data + subtypes
GEO_GSE110054_resistance_UP_genes.csv	Genes induced in resistance
GEO_GSE110054_resistance_DOWN_genes.csv	Genes repressed in resistance
GEO_GSE110054_kinase_suppression_vs_rna_rebound.csv	Suppression vs. rebound
nn_training_history.csv	Neural-network training history
nn_predictions_test.csv	Predictions on the test set